

Accounting and Accountability

Week 11



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Key concepts

Identify relevant information for decision-making

- Decisions made by businesses can have short-term effects or long-term impacts, or in some situations, both.
- Short-term decisions often address a temporary circumstance or an immediate need while long-term decisions align more with permanent problem solving and meeting strategic goals.
- These two types of decisions require different types of analyses and different types of accounting and non-accounting information.

Examples of Short-Term and Long-Term Business Decisions

Short-Term Business Decisions

- Accepting a special production order
- Determining the best product mix from current products
- Outsourcing a part or service
- Further processing or refining a current product

Long-Term Business Decisions

- Buying new equipment versus remodeling old equipment
- Choosing which products to manufacture
- Expanding into a new area or country
- Diversifying by buying another business

Key concepts

Relevant information for short-term decision-making

Business decision-making can be outlined as a process that is applied by management with each decision that is made. The process of decision-making in a managerial business environment can be summed up in these steps.

- 1. Identify the objective or goal. For a business, typically the goal is to maximise revenues or minimise costs.
- 2. Identify alternative courses of action that can achieve the goal or address an obstacle that is hindering goal achievement.
- 3. Perform a comprehensive analysis of potential solutions. This includes identifying revenues, costs, benefits, and other financial and qualitative variables.
- 4. Decide, based upon the analysis, the best course of action.
- 5. Review, analyse, and evaluate the results of the decision.

Key concepts

Relevant information for short-term decision-making

In carrying out step three of the managerial decision-making process, a **differential analysis** compares the relevant costs and revenues of potential solutions. First, it is important to understand that there are many types of short-term decisions that a business may face, but these decisions always involve choosing between alternatives. Examples of these types of decisions include:

- determining whether to accept a special order
- making a product or component versus buying the product or component
- performing additional processing on a product
- keeping versus eliminating a product or segment.

In each of these situations, the business should **compare the relevant costs and the relevant revenues** of one alternative to the relevant costs and relevant revenues of the other alternative(s). Therefore, an important step in the differential analysis of potential solutions is to identify the relevant costs and relevant revenues of the decision.

Key concepts

Identifying relevant revenues and costs

- In the context of decision-making, something is relevant if it will influence the decision being made.
- Looking only at these differences – of both costs and revenues – between the alternatives, is known as **differential analysis**.
- In conducting these types of analyses between alternatives, the initial focus will be on each quantitative factor of the analysis – in other words, the component that can be measured numerically.
- However, in decision-making, it is important also to consider each qualitative factor, which is one that cannot be measured numerically.

Key concepts

Avoidable versus unavoidable costs

Management must determine if a cost is avoidable or unavoidable because in the short run, only avoidable costs are relevant for decision-making purposes.

- An **avoidable cost** is one that can be eliminated (in whole or in part) by choosing one alternative over another.
- An **unavoidable cost** is one that does not change or go away in the short-run by choosing one alternative over another.
- Variable costs are avoidable costs, since variable costs do not exist if the product is no longer made, or if the portion of the business (such as a segment or division) that generated the variable costs ceases to operate.
- Fixed costs, on the other hand, may be unavoidable, partially unavoidable, or avoidable only in certain circumstances.

Key concepts

Avoidable versus unavoidable costs

Example: AlexCo's Wagons

AlexCo produces collapsible wagons that are popular with beachgoers, shoppers, gardeners and parents. Annual sales have been 100,000 wagons per year. The retail selling price of each wagon is \$67.00. To date, AlexCo has produced each of the components used in making the wagons but has been approached by DAL, Inc. with an offer to provide the axle and wheel assembly for \$18.75 per assembly. AlexCo's costs to produce the axle and wheel assembly are \$9.00 in direct materials, \$6.50 in direct labor, \$3.57 in variable overhead, and \$2.50 in fixed overhead. Twenty-five percent of the fixed overhead is avoidable if the assembly is produced by DAL. Should AlexCo continue to make the axle and wheel assembly or should it buy the assembly from DAL, Inc.?

Key concepts

	Relevant Costs	
	Make Internally	Buy from DAL, Inc.
Direct materials	\$ 9.00	
Direct labor	6.50	
Variable overhead	3.57	
Avoidable fixed costs	0.63	
Total unit relevant cost	19.70	\$ 18.75
Units required	100,000	100,000
Total relevant costs	\$1,970,000	\$1,875,000

Ignoring qualitative factors, it would be more cost effective for AlexCo to buy the axle and wheel assembly from DAL, Inc. However, AlexCo should be certain of any qualitative issues and not solely base their decision on the quantitative analysis. What is the quality of the axels and wheel assemblies from DAL? Will they deliver on time? Where are the parts made?

Key concepts

Sunk costs

- A sunk cost is one that cannot be avoided because it has already occurred.
- A sunk cost will not change regardless of the alternative that management chooses;
- therefore, sunk costs have no bearing on future events and are **not relevant** in decision-making.

For example, suppose you have an old car, a hand-me-down from your grandmother, and last year you spent \$1,600 on repairs and new tires and were just told by your mechanic that the car needs \$1,200 in repairs to operate safely. Your goal is to have a safe and reliable car. Your alternatives are to get the repairs completed or trade in the car for a newer used car. From a quantitative perspective, you have gathered the following information to help with your decision. The trade-in value of your old car will be the minimum given by the dealer, or \$200. The newer used car will require you to make monthly payments of \$150 for two years. In analysing your two alternatives, what costs do you consider?

Key concepts

Sunk costs

- Remember, the \$1,600 you have already spent (note the past tense) is a sunk cost; it is a consequence of a past decision.
- In this example, the relevant costs for each alternative are the following: \$1,200 in current repair costs to keep your current car or \$3,400 (from the 24 payments of \$150 minus \$200 for the trade in) to buy a newer used car.
- Obviously, you also would consider qualitative factors, such as the sentimental value of your grandmother's car, the excitement of having a newer car or the need for the reliability of a newer car.

Key concepts

Future costs that do not differ (irrelevant costs)

- Any future cost that does not differ between the alternatives is not a relevant cost for the decision.
- For example, if a company is considering baking either bagels or doughnuts and both baked goods require \$0.30 worth of flour, then the cost of flour would not be a relevant cost in determining which of the two had the highest production cost.

Key concepts

Opportunity costs

- you may be faced with opportunity costs, which are the costs associated with not choosing the other alternative. Opportunity costs are relevant.

Example: Sydney Sweets is a lolly and candy manufacturer in a resort town, just bought a new candy machine for \$27,000 and is planning to increase the production of their signature hard candy. Due to the increased production, Sydney Sweets is deciding between hiring two university students on a casual basis or one full-time employee. Each university student would work half days totalling 20 hours per week, and would earn \$22 per hour. The full-time employee would work full days 40 hours per week and would earn \$22 per hour plus the equivalent of \$2 per hour in sick and annual leave. Each employee is given two t-shirts to wear as their uniform. The t-shirts cost Sydney Sweets \$25 each. In addition, Sydney Sweets provides disposable hair coverings and plastic gloves for the employees. Each employee uses, on average, six sets of gloves per eight-hour shift or four sets per four-hour shift. One hair covering per shift per person is typical. The cost of the hair covering is \$0.05 per covering and the cost of a pair of gloves is \$0.02 per pair. Identify any relevant costs, relevant revenues, sunk costs, and opportunity costs that Carolina Clusters needs to consider in making the decision whether to hire two part-time employees or one full-time employee.

Item	University student option (2 students)	Full time employee option	Relevant or not relevant?
New candy machine	\$27000	\$27000	Not relevant
Hourly rate	\$22	\$22	Not relevant
Hours worked per week	20hrs x 2 = 40 hours	40 hours	Not relevant
Sick and annual leave	\$0	\$2 per hour x 40 hours = \$80	RELEVANT
Uniforms	2 x \$25 = \$50	1 x \$25 = \$25	RELEVANT
Gloves	2 shifts per day, 4 sets of gloves, \$0.02c per pair = \$0.16	1 shift per day x 6 pairs of gloves, \$0.02 per pair = \$0.12	RELEVANT
Hair covering	2 shifts x \$0.05 = \$0.10	1 shift x \$0.05 = \$0.05	RELEVANT
Revenues			No information given – not relevant
TOTAL RELEVANT COSTS	Sick/annual leave + Uniforms + gloves + hair covering = \$0 + \$50 + \$0.16 + \$0.10 = \$50.26	Sick/annual leave + Uniforms + gloves + hair covering = \$80 + \$25 + \$0.12 + \$0.05 = \$105.17	

Sunk costs: \$27,000 for the candy machine – this does not change based on the employees
Opportunity costs: None

Key concepts

Relevant costs

What are relevant costs and revenues?	What are NOT relevant costs and revenues?
<i>Different</i> for each choice. Future costs and revenues that are different for each option.	The <i>same</i> for each choice. Future costs and revenues that are the <i>same</i> for each option.
<i>Avoidable</i> - these are usually the variable costs. Sometimes the fixed cost will be avoidable if they change depending on the decision. Make sure you check the question for extra information about fixed costs.	<i>Unavoidable</i> – Fixed costs that don't change no matter what choice is made are unavoidable. For example factory rent on the factory is the same no matter what decision is made, is unavoidable.
	<i>Sunk</i> costs – costs that you have already paid in the past. They are not relevant because they have already happened.
<i>Opportunity</i> costs – costs or revenues related to not choosing the other option.	

Quick Check

I have a big decision to make. I want to buy a puppy. Which puppy should I buy? I am considering a cavoodle which costs \$5,000 or a king charles cavalier which costs \$3,000. The vet costs will be about \$500 per year. The cavoodle will need to go to the groomer every month which will cost \$80 per month. Food costs will be approximately \$50 per month. The king charles cavalier will shed hair all over the house so I will need a new vacuum cleaner that will cost \$1,000. The cavoodle does not shed hair. I will also need to go to the pet shop and buy some supplies like a lead, a bed and some toys. I plan to spend about \$500 on these items.

The decision is which dog to buy. Which cost is relevant to this decision?

The cost to buy the dog.

The vet costs.

The food costs.

Pet shop supplies.

Quick Check

We will need to make a decision about whether to purchase new machinery or repair the old one. The old machinery was purchased five years ago for \$200,000. It can be sold today for approximately \$20,000. The cost to upgrade the machine so it works more efficiently is \$150,000. The cost of a new machine is \$250,000.

Which cost is not relevant?

The cost of the old machinery \$200,000.

Current sales price of the old machinery \$20,000.

Cost to upgrade \$150,000

Cost of the new machine, \$250,000

Key concepts

Accept or reject a special order

- Both manufacturing and service companies often receive requests to fill special orders. These special orders are typically for goods or services at a reduced price and are usually a one-time order that, in the short-run, does not affect normal sales.
- the management must consider several factors:
 1. The capacity required to fulfill the special order
 2. Whether the price offered by the buyer will cover the cost of producing the products
 3. The role of fixed costs in the analysis
 4. Qualitative factors

Key concepts

Accept or reject a special order

- **Example – Franco Pty Ltd**

Franco Pty Ltd, produces dental office examination chairs. Franco has the capacity to produce 5,000 chairs per year and currently is producing 4,000. Each chair retails for \$2,800, and the costs to produce a single chair consist of direct materials of \$750, direct labour of \$600, and variable overhead of \$300. Fixed overhead costs of \$1,350,000 are met by selling the first 3,000 chairs. Franco has received a special order from Ghanem, Inc., to buy 800 chairs for \$1,800. Should Franco accept the special order?

Key concepts

Accept or reject a special order

	Current Cost to Produce	Special Order Price Offer	Difference in Favor of Accepting Special Order
Direct materials	\$ 750		
Direct labor	600		
Variable overhead	300		
Variable costs to produce	<u>1,650</u>		
Special offer price		\$1,800	\$150 per chair

Since Franco has already met his fixed costs with current production and since he has the capacity to produce the additional 800 units, Franco only needs to consider his variable costs for this order. Franco's variable cost to produce one chair is \$1,650. Ghanem is offering to buy the chairs for \$1,800 apiece. By accepting the special order, Franco would meet his variable costs and make \$150 per chair. Considering only quantitative factors, Franco should accept the special offer.

Key concepts

How would Franco's decision change if the factory was already producing at capacity at the time of the special offer?

Assume that, to fill the order from Ghanem, Franco would have to run an extra shift, and this would require him to hire a temporary production manager at a cost of \$90,000. Assume no other fixed costs would be incurred. Also assume Franco will incur additional costs related to maintenance and utilities for this extra shift and estimates those costs will be \$70,000.

	Current Production	Special Order Current Offer
Selling price	\$2,800	\$1,800
Variable cost to produce	<u>1,650</u>	1,650
Additional costs to recover*		<u>200</u>
Contribution margin	\$1,150	\$ (50)
*\$90,000 supervisor salary + \$70,000 additional costs = \$160,000 in costs to recover ÷ 800 chairs in special order = \$200 per chair additional costs due to capacity issue.		

Quick Check

LOB can make up to 300 bags per month. They are currently making and selling 270 per month for \$150.

They get a special order from a customer who wants to purchase 50 bags for \$80 each.

Should they accept or reject the special order?

Accept the special order.

Reject the special order.

I don't know.

Quick Check

LOB can make and sell up to 400 bags per month and is currently making and selling 250 bags. A customer has asked for a special order of 100 bags and will pay \$3,000 for them all. The normal business costs are:

Variable costs per bag include leather \$15, labour \$10, overhead \$5 and selling and administration of \$2.

Fixed costs per month are \$15,000 and include factory rent and other fixed factory costs, manager's salary, website costs and social media marketing.

If the business accepts the special order they will only pay half the selling and administration variable cost. The fixed costs are will not change if the business accepts the special order.

Accept the special order.

Reject the special order.

I don't know.

Key concepts

Make or buy a component

- the make-versus-buy decision involves both quantitative and qualitative analysis.
- The quantitative component requires cost analysis to determine which alternative is more cost effective.
- This cost analysis can be performed by looking at the cost to buy the component versus the cost to produce the component, which allows us to make a decision based on an analysis of unavoidable costs.
- For example, the costs to produce will include direct materials, direct labor, variable overhead, and fixed overhead. If the business chooses to buy the component instead, the avoidable costs will go away but unavoidable costs will remain and would need to be considered as part of the cost to buy the component.

Key concepts

Make or buy a component

- Thermal Mugs, Inc., manufactures various types of leak-proof personal drink carriers. Thermal's T6 container, its most insulated carrier, maintains the temperature of the liquid inside for 6 hours. Thermal has designed a new lid for the T6 carrier that allows for easier drinking and pouring. The cost to produce the new lid is \$2.19:

Direct materials	\$0.87
Direct labor	0.45
Fixed overhead	0.51
Variable overhead	<u>0.36</u>
Total unit cost	\$2.19

- Plato Plastics has approached Thermal and offered to produce the 120,000 lids Thermal will require for current production levels of the T6 carrier, at a unit price of \$1.75 each. Is this a good deal? Assume all the fixed costs are not tied directly to the production of the lid and therefore will still exist even if the lid is purchased externally from Plato.

Key concepts

Make or buy a component

	Relevant Costs	
	Make Internally	Buy from Plato
Direct materials	\$ 0.87	
Direct labor	0.45	
Variable overhead	0.36	
Total unit relevant cost	1.68	\$ 1.75
Units required	120,000	120,000
Total relevant costs	\$201,600	\$210,000

Calculations show that when the relevant costs are compared between the two alternatives, it is more cost effective for Thermal to produce the 120,000 units of the T6 lid internally than to purchase it from Plato.

Key concepts

Make or buy a component

Suppose that, of the \$0.51 in fixed costs per unit of the T6 lid, \$0.12 of those fixed costs are associated with interest costs and insurance expenses and thus would be avoidable if the T6 lid is purchased externally rather than produced internally. How does that change the analysis?

	Relevant Costs	
	Make Internally	Buy from Plato
Direct materials	\$ 0.87	
Direct labor	0.45	
Variable overhead	0.36	
Avoidable fixed costs	0.12	
Total unit relevant cost	<u>1.80</u>	\$ 1.75
Units required	<u>120,000</u>	<u>120,000</u>
Total relevant costs	\$216,000	\$210,000

Key concepts

Make or buy a component

These analyses only considered the quantitative factors in a make-versus-buy decision, but there are qualitative factors to consider as well, including:

- Will the T6 lid made by Plato meet the quality requirements of Thermal?
- Will Plato continue to produce the T6 lid at the \$1.75 price, or is this a teaser rate to obtain the business, with the plan for the rate to go up in the future?
- Can Plato continue to produce the quantity of the lids desired? If more or fewer are needed from Plato, is the adjusted production level obtainable, and does it affect the cost?
- Does using Plato to produce the lids displace Thermal workers or hamper morale?
- Does using Plato to produce the lids affect the reputation of Thermal?
- Buying the lid likely means that Thermal has excess production capacity that can now be applied to making other products.

Key concepts

Make or buy a component

Advantages of Outsourcing

- Utilises external expertise, removes the need for in-house expertise
- Frees up capacity for other uses
- Frees up capital for other uses
- Allows management to focus on competitive strengths
- Transfers some production and technological risks to supplier

Disadvantages of Outsourcing

- Takes away control over quality and timing of production
- May limit ability to upsize or downsize production
- May have hidden costs and/or a lack of stability of price
- May diminish innovation
- Often makes it difficult to bring the production back in-house once it has been removed

Quick Check

LOB is considering adding a new product, leather wallets.

The cost to make 100 leather wallet includes:

Leather wallets	Total	Per unit
Direct materials	\$500	\$5
Direct labour	\$500	\$5
Variable overhead	\$200	\$2
Fixed overhead	\$500	\$5
Total	\$1,700	\$17

LOB can buy a wallet for \$15 per unit.

If LOB purchases the wallets, they will still have to pay the fixed overhead. The fixed overhead is unavoidable.

What is the relevant cost per unit to make a wallet?

I don't know.

\$15

\$17

\$12✓

Quick Check

LOB is considering adding a new product, leather wallets.

The cost to make 100 leather wallet includes:

Leather wallets	Total	Per unit
Direct materials	\$500	\$5
Direct labour	\$500	\$5
Variable overhead	\$200	\$2
Fixed overhead	\$500	\$5
Total	\$1,700	\$17

LOB can buy a wallet for \$15 per unit.

If LOB purchases the wallets, they will still have to pay the fixed overhead. The fixed overhead is unavoidable.

Based on your analysis of the last question, should LOB make or buy the wallets?

Buy.

I don't know.

Make.✓

Quick Check

LOB is considering adding a new product, leather wallets.

The cost to make 100 leather wallet includes:

Leather wallets	Total	Per unit
Direct materials	\$500	\$5
Direct labour	\$500	\$5
Variable overhead	\$200	\$2
Fixed overhead	\$500	\$5
Total	\$1,700	\$17

LOB can buy a wallet for \$15 per unit.

If LOB purchases the wallets, they will still have to pay the fixed overhead. The fixed overhead is unavoidable.

If LOB buys the wallets they can now rent out part of their factory (the space they would have used to make the wallets) and earn \$500 in rental income.

Should LOB make or buy the wallets?

Make.

Buy. ✓

I don't know.

Key concepts

Keep or discontinue a segment/division or product/service

A segment is a portion of the business that management believes has sufficient similarities in product lines, geographic locations, or customers to warrant reporting that portion of the company as a distinct part of the entire company.

- Two basic approaches can be used to analyse data in this type of decision.
 1. Compare contribution margin and fixed costs. In this method, the contribution margins with and without the segment (or division or product line) are determined.
 2. Compare total net income if the segment/product/service is retained vs dropped.

Key concepts

Example-SnowBucks

Suppose SnowBucks, Inc., has three product lines: snow boots, snow sporting equipment, and a clothing line for winter sports. It has been brought to senior management's attention that the snow boot product line is unprofitable. The table below shows the data presented to senior management:

	Snow Boots	Snow Sporting Equipment	Clothing Line	Total
Sales	\$1,150,000	\$1,540,000	\$1,354,000	\$4,044,000
Cost of goods sold				
Variable manufacturing expenses	423,000	507,000	378,000	1,308,000
Fixed manufacturing expenses	392,000	413,000	353,000	1,158,000
Gross margin	335,000	620,000	623,000	1,578,000
Selling and administrative expenses				
Variable selling and administrative expenses	195,000	130,000	147,000	472,000
Fixed selling and administrative expenses	216,000	216,000	216,000	648,000
Operating income	<u>\$ (76,000)</u>	<u>\$ 274,000</u>	<u>\$ 260,000</u>	<u>\$ 458,000</u>

Key concepts

Example-SnowBucks

In conducting the analysis, the accounting team discovers that each product line is allocated certain costs over which the product line managers have no control. These allocated costs are typically associated with areas of the company that do not generate revenue but are necessary for the running of the business, such as salaries for executives, human resources, and accounting at headquarters.

Companies often allocate these costs to other parts of the business based on some formula, such as dividing the total costs by the number of divisions or segments, as percentage of total revenue, or as percentage of total square footage.

SnowBucks currently allocates these costs equally to the three product lines, and all the fixed selling and administrative expenses are considered allocated costs. In addition, the fixed manufacturing expenses represent factory rent, depreciation, and insurance, and all these costs will continue to exist regardless of whether the snow boot division continues. However, included in the fixed manufacturing expenses is the \$75,000 salary of a sales supervisor for each division. This is an avoidable fixed cost as this cost would no longer exist if any division ceased operating.

Key concepts

SnowBucks worked example – using the contribution margin method

	Snow Boots	Snow Sporting Equipment	Clothing Line	Total
Sales	\$1,150,000	\$1,540,000	\$1,354,000	\$4,044,000
Variable expenses				
Variable manufacturing expenses	423,000	507,000	378,000	1,308,000
Variable selling and administrative expenses	<u>195,000</u>	<u>130,000</u>	<u>147,000</u>	<u>472,000</u>
Contribution margin	532,000	903,000	829,000	2,264,000
Direct fixed manufacturing expenses	<u>75,000</u>	<u>75,000</u>	<u>75,000</u>	<u>225,000</u>
Product margin	457,000	828,000	754,000	2,039,000
Allocated fixed expenses				
Fixed selling and administrative expenses				648,000
Fixed manufacturing expenses				<u>933,000</u>
Operating income				<u><u>\$ 458,000</u></u>

This new analysis shows that when the relevant costs and revenues are considered, it is apparent the snow boot product line is contributing toward meeting the fixed costs of the business and therefore to overall corporate profitability. The reason the snow boot product line was showing an operating loss was due to the allocation of common costs.

Key concepts

Making decisions when resources are scarce or limited

Understanding the importance of capacity

- Companies use various resources to be productive. These resources, which include time, labour, space, and machines, are limited, thus constraining the ability of a company to have unlimited productive capacity.
- These constraints require companies to make decisions on the best ways to allocate their resources in a way that maximises the benefit to the firm.
- The question as to which products and how many should be made is a common constraint problem.
- The initial step in allocating scarce or constrained resources is to determine the unit contribution margin, which is the selling price per unit minus the variable cost per unit, for each product or service.
- The total contribution margin will be maximised by promoting those products or accepting those orders with the highest contribution margin in relation to the scarce resource.

Key concepts

Example – Wood World

Wood World, Inc., produces wooden desks, chairs, and bookcases. These items are produced using the same machines, and there is a maximum of 80,000 machine-hours available during the year. Wood World is limited in producing its products by the number of possible machine-hours. Orders have been received for 60,000 desks, 48,000 chairs, and 40,000 bookcases, which will require 94,000 machine-hours to produce. Since there are not enough machine-hours available to fill all of the orders, which orders should Wood World fill first?

	Desks	Chair	Bookcase
Hours to produce	1	0.50	0.25
Selling price	\$ 350	\$ 200	\$ 175
Direct materials	\$ 40	\$ 30	\$ 35
Direct labor	\$ 70	\$ 65	\$ 50
Variable overhead	\$ 55	\$ 50	\$ 45
Fixed overhead	\$ 28	\$ 32	\$ 24

Key concepts

Example – Wood World

To address this question, Wood World must find the contribution margin per machine-hour since machine-hours are the constraining factor for production.

	Desks	Chair	Bookcase
Selling price	\$ 350	\$ 200	\$ 175
Direct materials	\$ 40	\$ 30	\$ 35
Direct labor	\$ 70	\$ 65	\$ 50
Variable overhead	\$ 55	\$ 50	\$ 42
Contribution margin	\$ 185	\$ 55	\$ 48
Hours to produce	1	0.50	0.25
Contribution margin per machine-hour	\$ 185	\$ 110	\$ 192

Key concepts

Example – Wood World

Wood World should fulfill the orders for bookcases first, desks second, and chairs last.
How many of each item will be produced?

Available machine hours	80,000
Hours to fill bookcase orders ($40,000 \times 0.25$)	<u>10,000</u>
Remaining hours	70,000
Hours to fill desk orders ($60,000 \times 1$)	<u>60,000</u>
Remaining hours	10,000
Hours needed to produce chairs	<u>$\div 0.50$</u>
Chair orders that would be filled	20,000

Key concepts

Increasing the capacity of the constraint or **bottleneck** is also called **relaxing the constraint or elevating the constraint**. Some specific examples of ways to relax the constraint include:

- Keep the production facilities open longer hours. This may allow the work-flow through the bottleneck area to be slowed and thus prevent the bottleneck from occurring. However, this may require paying workers overtime pay.
- If working extra hours is not a viable option, then moving additional workers to the bottleneck area may be beneficial as long as the areas from which they were moved are adequately covered and additional problem areas do not result.
- Instead of using current workers, additional staff may be hired to smooth the work flow through the bottleneck area.
- Outsource some or all of the work in the area of the bottleneck. It may be cheaper and more cost effective to buy parts of components than to slow production due to the bottleneck.
- Redesign the production process to prevent the bottleneck by adding more resources to eliminate the bottleneck, re-organising the process to distribute the bottleneck-causing activities to different parts of the production process, or managing processing times at other stages prior to the bottleneck to help prevent the bottleneck from occurring.
- Insuring a minimal number of defects and rework, since they typically slow the production process and thus add to the bottleneck.

Quick Check

Colonial Heritage makes reproduction colonial furniture from select hardwoods.

	<i>Chairs</i>	<i>Tables</i>
Selling price per unit	\$80	\$400
Variable cost per unit	\$30	\$200
Board feet per unit	2	10
Monthly demand	600	100

The company's supplier of hardwood will only be able to supply 2,000 board feet this month. Is this enough hardwood to satisfy demand?

- a. Yes**
- b. No**

Quick Check

Colonial Heritage makes reproduction colonial furniture from select hardwoods.

	<i>Chairs</i>	<i>Tables</i>
Selling price per unit	\$80	\$400
Variable cost per unit	\$30	\$200
Board feet per unit	2	10
Monthly demand	600	100

The company's supplier of hardwood will only be able to supply 2,000 board feet this month. Is this enough hardwood to satisfy demand?

a. Yes

b. No

$$(2 \times 600) + (10 \times 100) = 2,200 > 2,000$$

Quick Check

	<i>Chairs</i>	<i>Tables</i>
Selling price per unit	\$80	\$400
Variable cost per unit	\$30	\$200
Board feet per unit	2	10
Monthly demand	600	100

The company's supplier of hardwood will only be able to supply 2,000 board feet this month. What plan would maximize profits?

- a. 500 chairs and 100 tables**
- b. 600 chairs and 80 tables**
- c. 500 chairs and 80 tables**
- d. 600 chairs and 100 tables**

Quick Check

	<i>Chairs</i>	<i>Tables</i>	
Selling price per unit	<i>Chairs</i>	<i>Tables</i>	
Variable cost per unit	Selling price	\$ 80	\$ 400
Board feet per unit	Variable cost	30	200
Monthly demand	Contribution margin	\$ 50	\$ 200
	Board feet	2	10
	CM per board foot	\$ 25	\$ 20
	Production of chairs	600	
	Board feet required	1,200	
	Board feet remaining	800	
	Board feet per table	10	
	Production of tables	80	

The company's supplier of hardboard has informed the company that only 800 board feet are available for use in the production of chairs and tables this month. What plan would maximize the company's contribution margin?

- a. 500 chairs and 100 tables
- b. 600 chairs and 80 tables**
- c. 500 chairs and 80 tables
- d. 600 chairs and 100 tables

Quick Check

As before, Colonial Heritage's supplier of hardwood will only be able to supply 2,000 board feet this month. Assume the company follows the plan we have proposed. Up to how much should Colonial Heritage be willing to pay above the usual price to obtain more hardwood?

- a. \$40 per board foot**
- b. \$25 per board foot**
- c. \$20 per board foot**
- d. Zero**

Quick Check

As before, Colonial Heritage's supplier of hardwood will only be able to supply 2,000 board feet this month. Assume the company follows the plan we have proposed. Up to how much should Colonial Heritage be willing to pay above the usual price to obtain more hardwood?

- a. \$40 per board foot
- b. \$25 per board foot
- ☒ c. \$20 per board foot
- d. \$15 per board foot

The additional wood would be used to make tables. In this case, each board foot of additional wood will allow the company to earn an additional \$20 of contribution margin and profit.

Tutorial activities--Tutorial 11A

Prepare for the final exam

Tutorial activities--Tutorial 11A

Which items below are *not* relevant costs for decision making? There is more than one correct answer.

- a. Unavoidable fixed costs.
- b. Avoidable fixed costs
- c. Opportunity costs.
- d. Qualitative factors.
- e. Sunk costs.

Tutorial activities--Tutorial 11A

Here is the financial information for the corporate gift baskets for Marketing Love♥ (ML). A customer wants to buy a large special order of 2,000 baskets at a special price of \$40. What is the increase/decrease in profit if we accept this special order?

	Standard
Sales price	\$100
Direct material per unit	24
Direct labour per unit	6
Variable overhead per unit	6
Fixed overhead per unit	5

Increase in revenues from this special order:

Increase costs from this special order:

Increase in profit =

Tutorial activities--Tutorial 11A

Should ML accept this special order?

- a)Accept the special order.
- b)Reject the special order.
- c)I don't know.

What qualitative factors should ML consider in making this decision?

- Consider what would happen if existing customers found out about this order.
- We might also want to consider the long-term consequences of offering this special price, will this customer continue to ask for this special price.
- Are we able to satisfy all our existing customers along with this special order customer at our usual quality? Do we have the capacity to do this job.
- Who are you selling the special order to? Is it someone you want to build a new relationship with or is it an existing customer that you have a relationship with. Is the special order for a good cause or charity that you want to associate with.

If ML was already at maximum capacity, should they accept or reject the special order?

- a)Accept the special order.
- b)Reject the special order.
- c)I don't know.

Tutorial activities--Tutorial 11A

ML also makes a deluxe corporate gift basket which has the following financial information:

	Deluxe
Sales price	\$160
Direct material per unit	44
Direct labour per unit	8
Variable overhead per unit	6
Fixed overhead	10

A factory has offered to make and sell us 100 *deluxe* baskets. It would cost us \$65 per unit to buy from this factory. The fixed costs would remain the same if we decide to buy these products. Should we make or buy the deluxe basket?

Tutorial activities--Tutorial 11A

ML also makes a deluxe corporate gift basket which has the following financial information:

	Deluxe
Sales price	\$160
Direct material per unit	44
Direct labour per unit	8
Variable overhead per unit	6
Fixed overhead	10

In the ML deluxe basket make or buy decision, if we buy the deluxe basket then we can rent out part of our factory for \$2,000. Should we make or buy?

Cost to make the products

Cost to buy

In an outsourcing decision, what are some of the qualitative factors that need to be considered?

-
-

Tutorial activities--Tutorial 11B

ML has three types of baskets. They are wondering if it is profitable to continue to sell all these products or if they should only focus on one or two. The total fixed costs have been split between the three products. The total fixed cost will not change if they drop a product. Should ML drop the standard product?

	Basic	Standard	Deluxe
Sales revenue (100 units each)	\$8,000	\$10,000	\$16,000
Variable costs	3,300	3,700	4,900
Contribution margin	4,700	6,300	11,100
Fixed costs	4,000	8,100	9,000
Net profit	700	(1,800)	2,100

If ML drops the standard profit, what will be the increase or decrease in total profit of the business?

Tutorial activities--Tutorial 11B

ML makes three different baskets. Here is the financial information for the current products sold:

	Basic	Standard	Deluxe
Sales price	\$80	\$100	\$160
Direct material per unit	20	24	44
Direct labour per unit	5	6	8
Variable overhead per unit	4	6	6
Fixed overhead per unit	3	5	10

If the factory has no labour or machine hour constraints, which product or mix of products should the business make?

	Basic	Standard	Deluxe
Selling price	\$80	\$100	\$160
- Variable costs	29	36	58
= Contribution margin per product	51	64	102



Tutorial activities--Tutorial 11B

ML makes three different baskets. Here is the financial information for the current products sold:

	Basic	Standard	Deluxe
Sales price	\$80	\$100	\$160
Direct material per unit	20	24	44
Direct labour per unit	5	6	8
Variable overhead per unit	4	6	6
Fixed overhead per unit	3	5	10

Would the answer to last question be different if there was a constraint in relation to labour hours available?

	Basic	Standard	Deluxe
Selling price	\$80	\$100	\$160
- Variable costs	29	36	58
= Contribution margin per product	51	64	102
Labour hours per product	0.1 hours	0.2 hours	1 hour
Contribution margin per hour	$51/0.1 = 510$	$64/0.2 = 320$	$102/1 = 102$



Tutorial problems week 11

None.